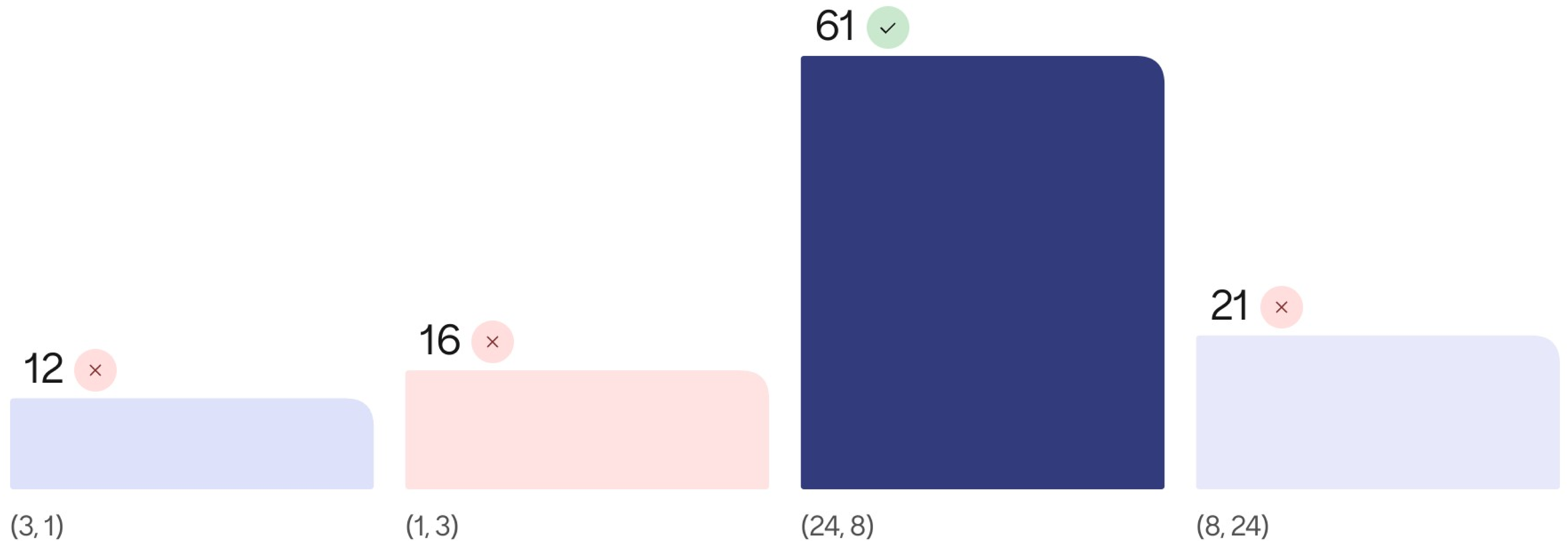
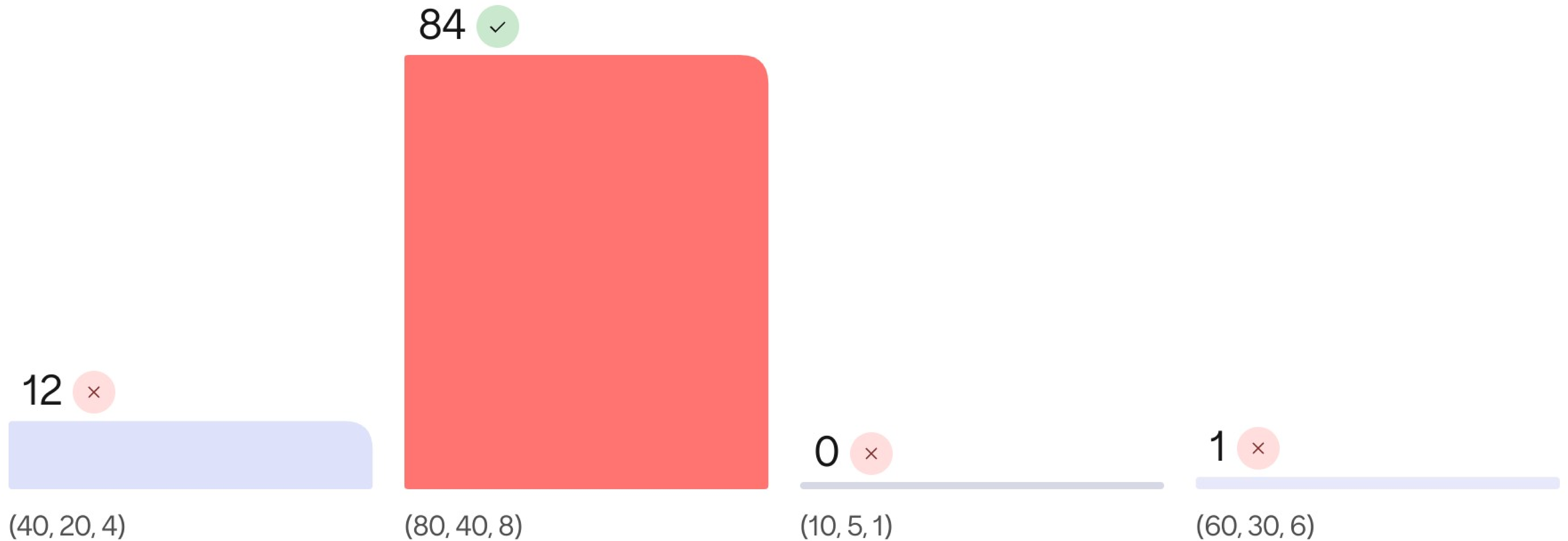


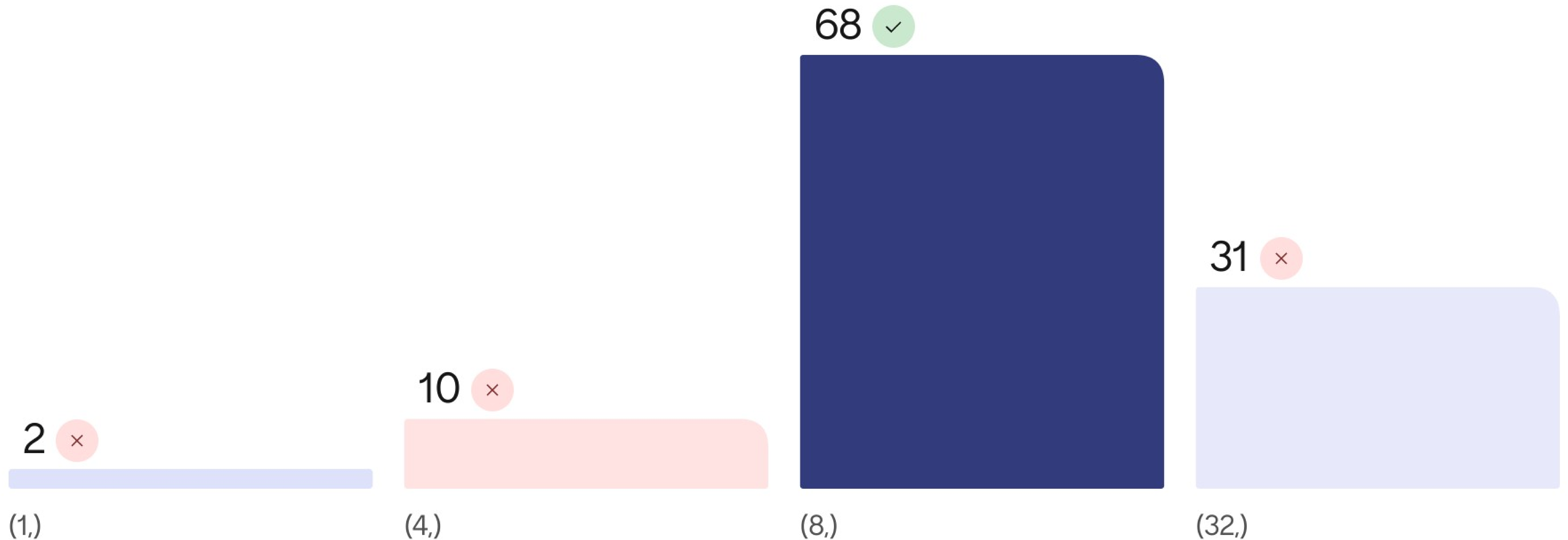
Let `a = np.ones((2, 3))`. What is `a.strides`?



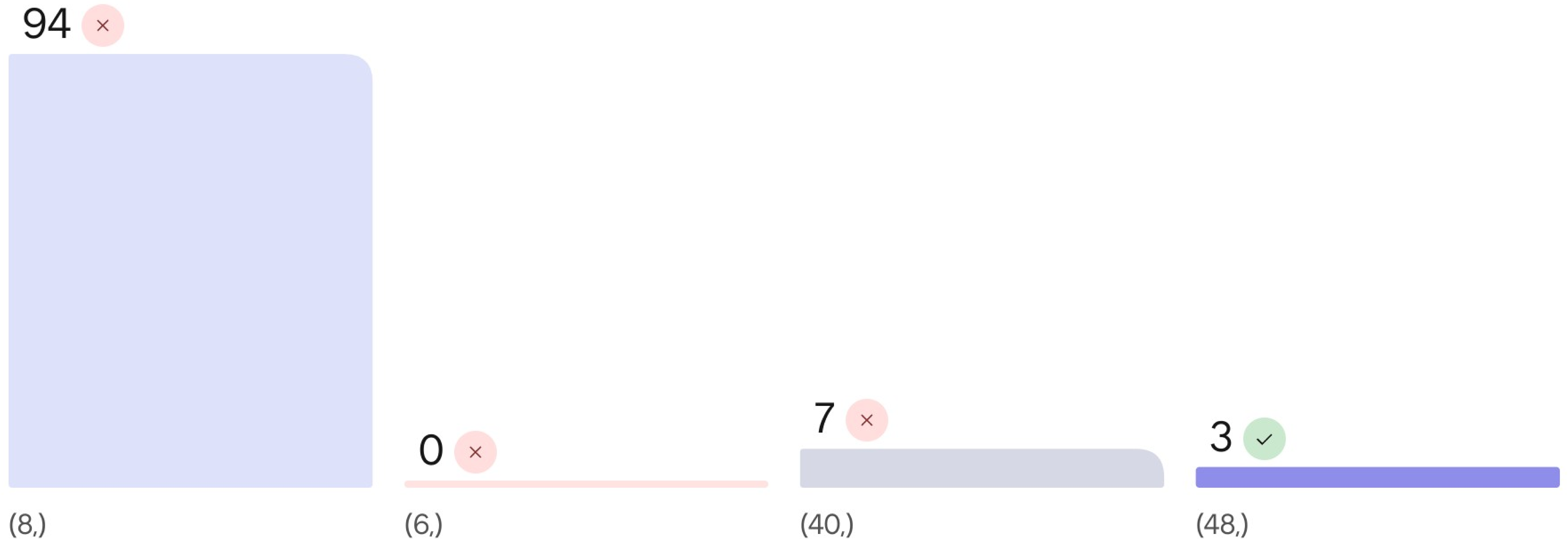
Let `a = np.ones((4, 2, 5))`. What is `a.strides`?



Let $a = \text{np.ones}(3, 4)$ and $b = a[1, :]$. What is $b.\text{strides}$?

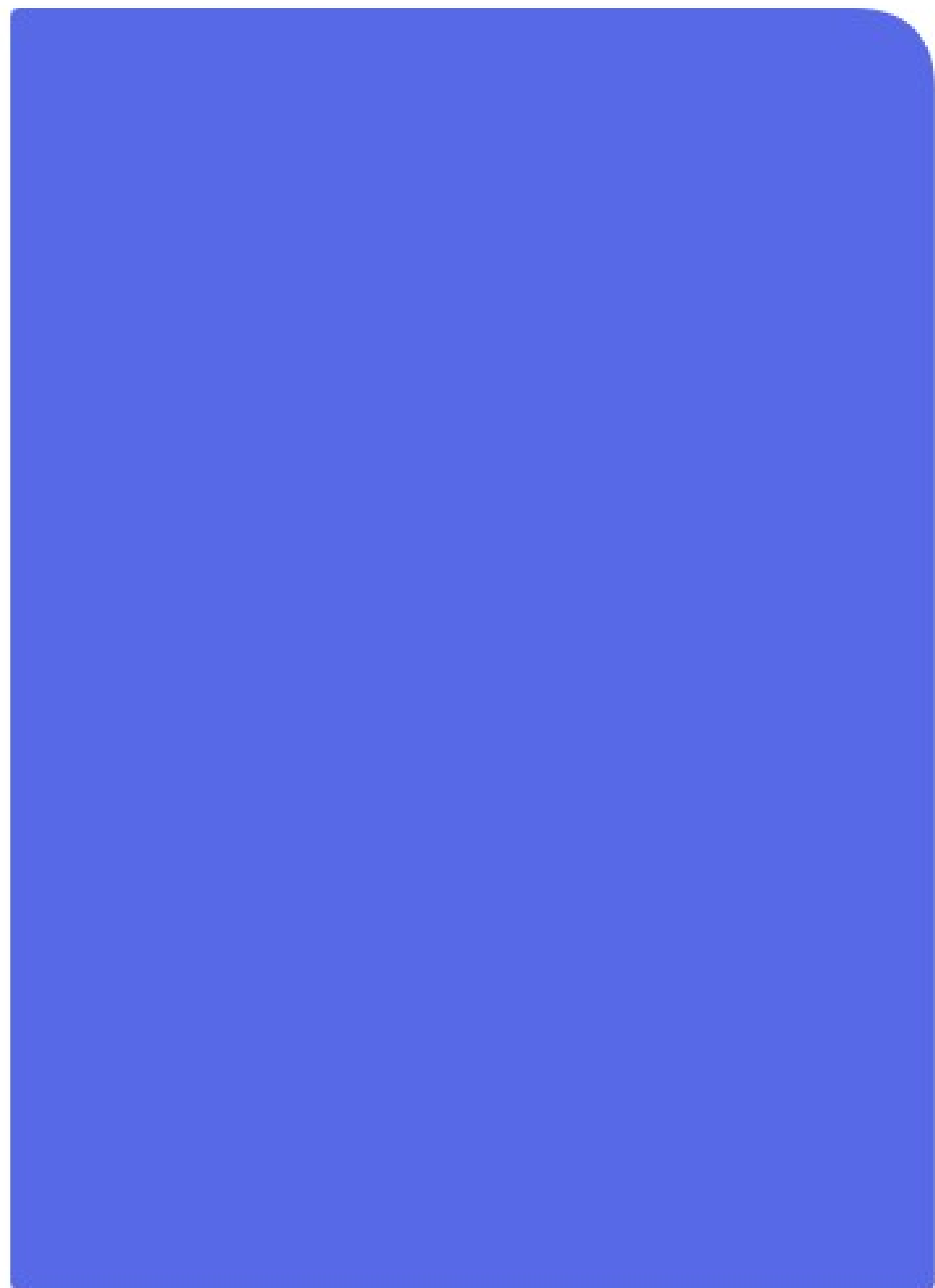


Let $a = \text{np.ones}(5, 6)$ and $b = a[1, :]$. What is $b.\text{strides}$?

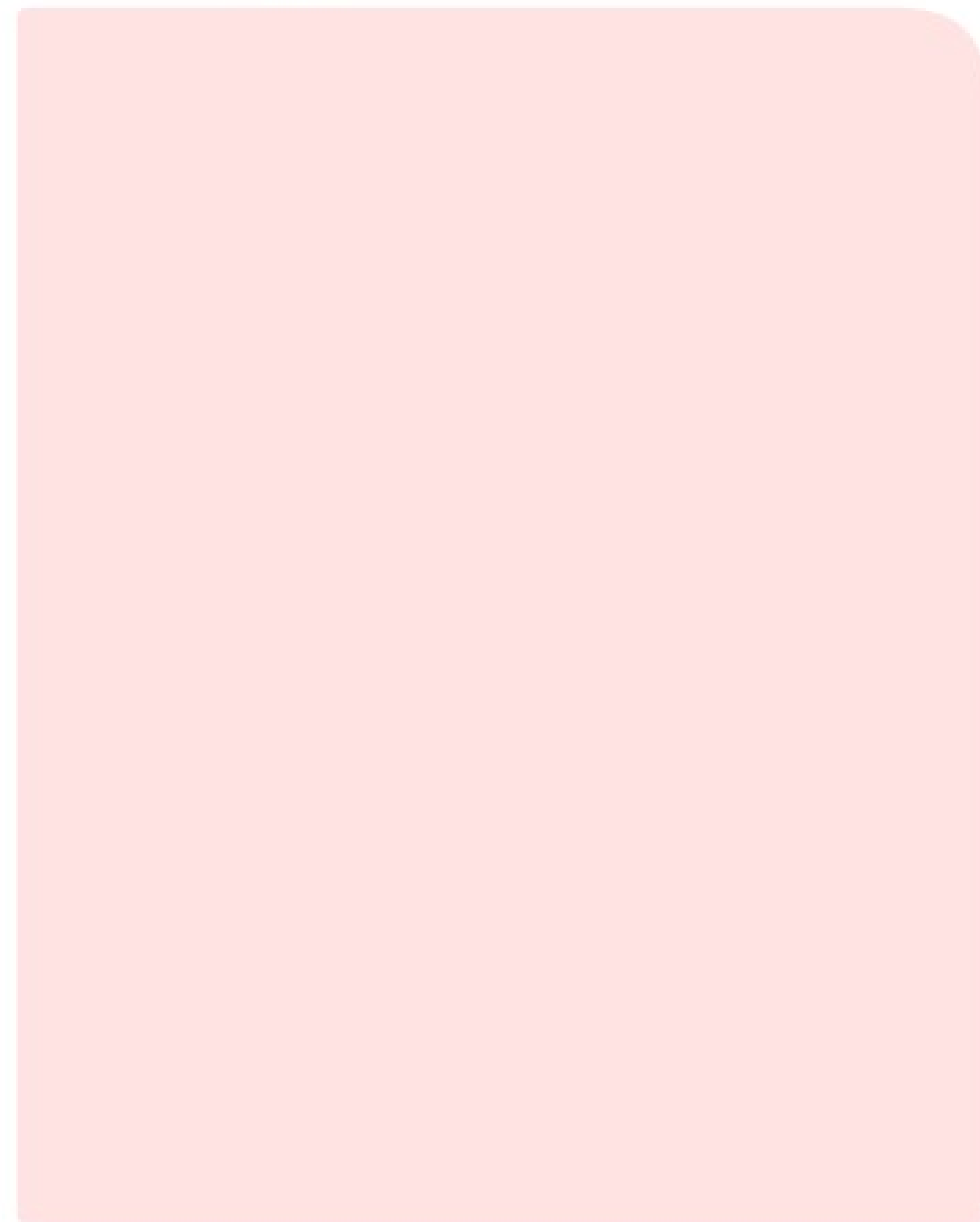


Let $a.shape = (2, 3)$. What will flip it horizontally?

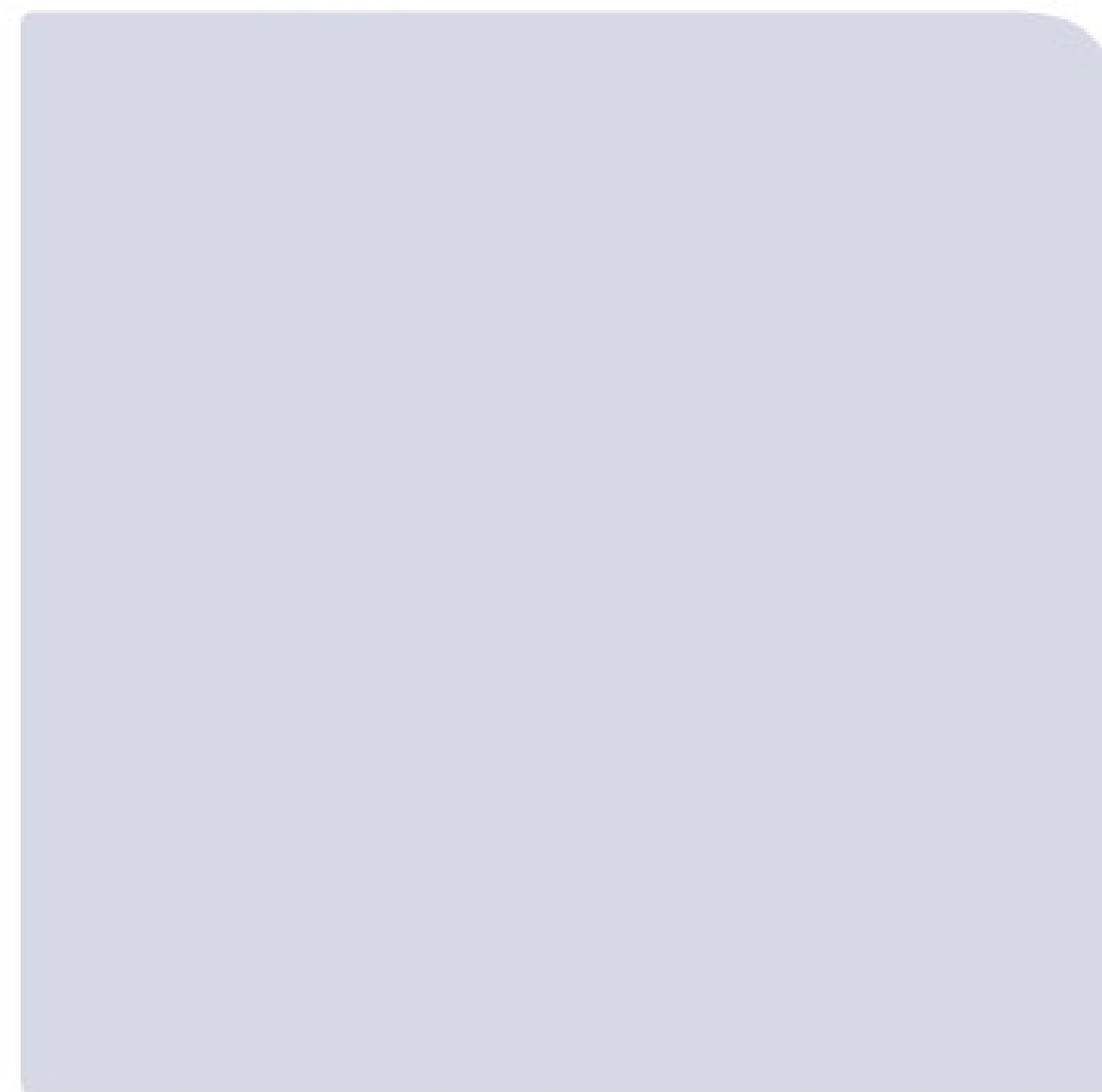
41 ✓

 $a[:, ::-1]$

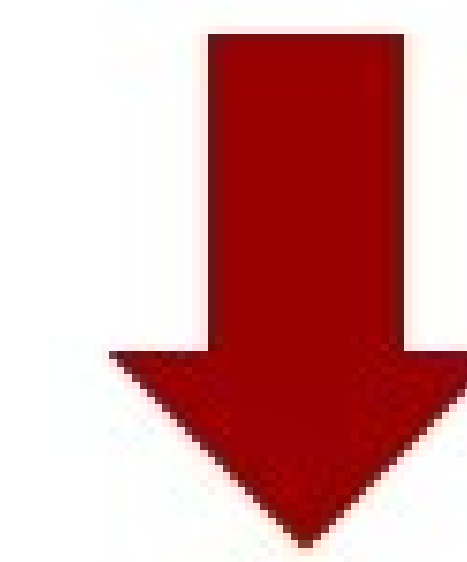
37 ✗

 $a[::-1, :]$

29 ✗

 $a[::-1, ::-1]$

1	2	3
4	5	6



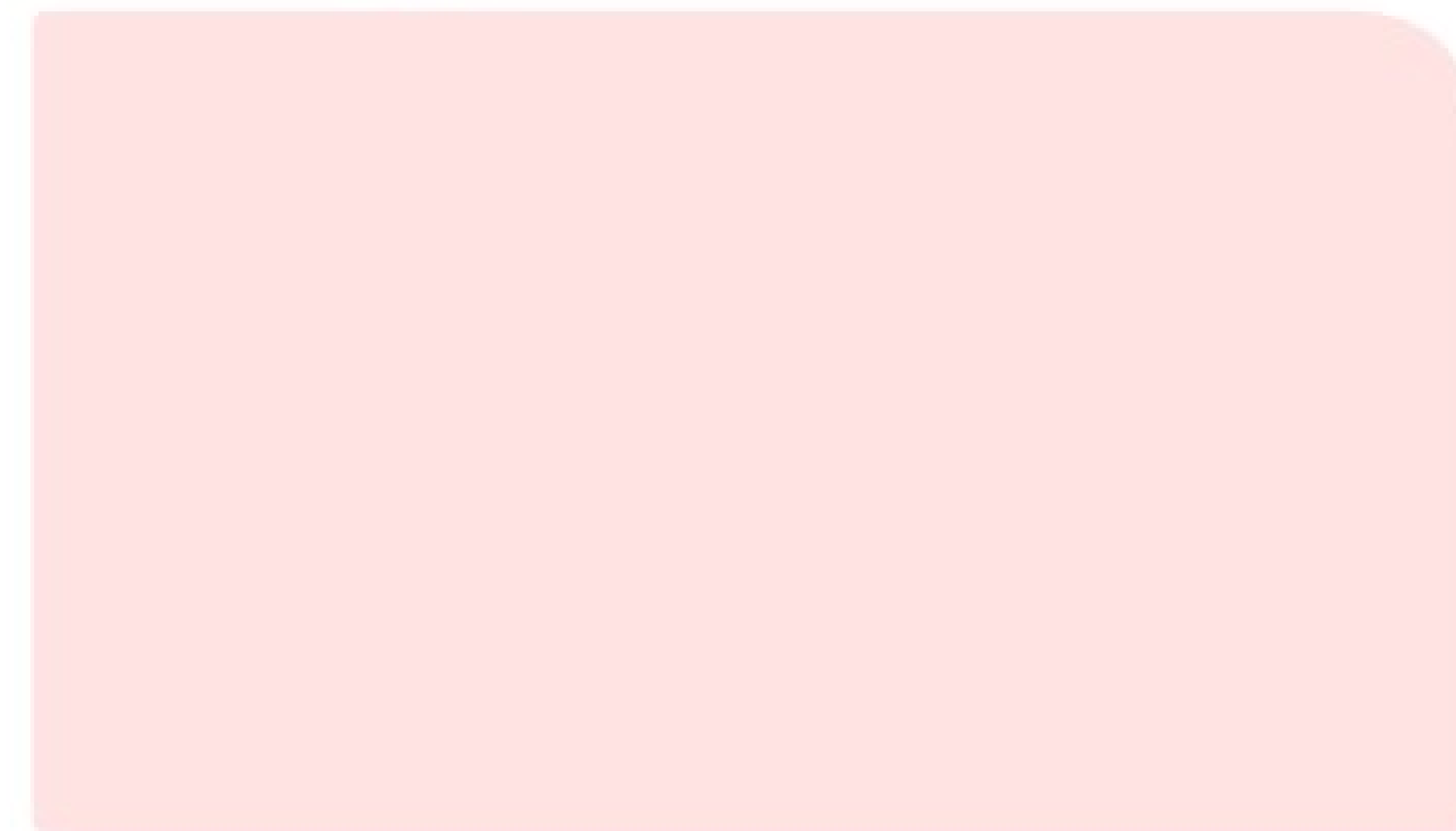
3	2	1
6	5	4

Let $a.\text{shape} = (N, M)$. What will rotate it 90 degrees?



$a[::-1, ::-1]$

29 ✖



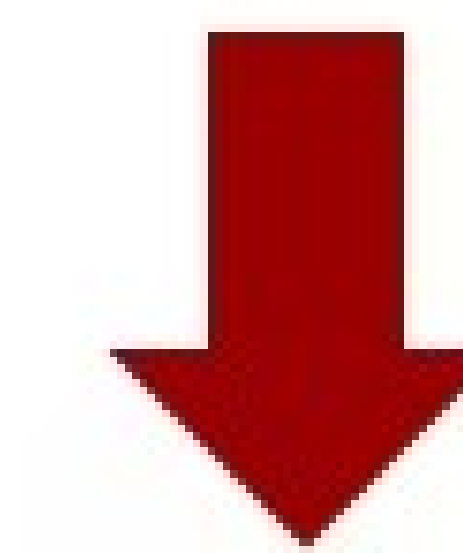
$a.T[::-1]$

72 ✔



$a.T[:, ::-1]$

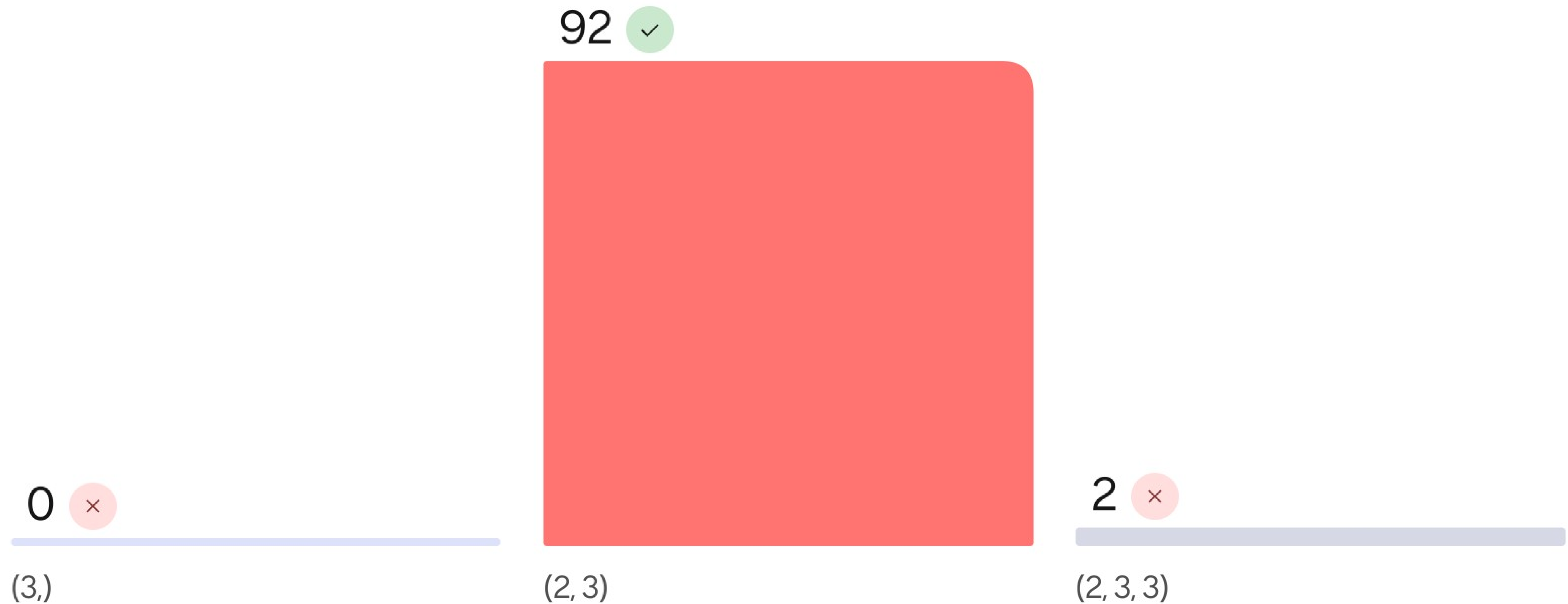
1	2	3
4	5	6



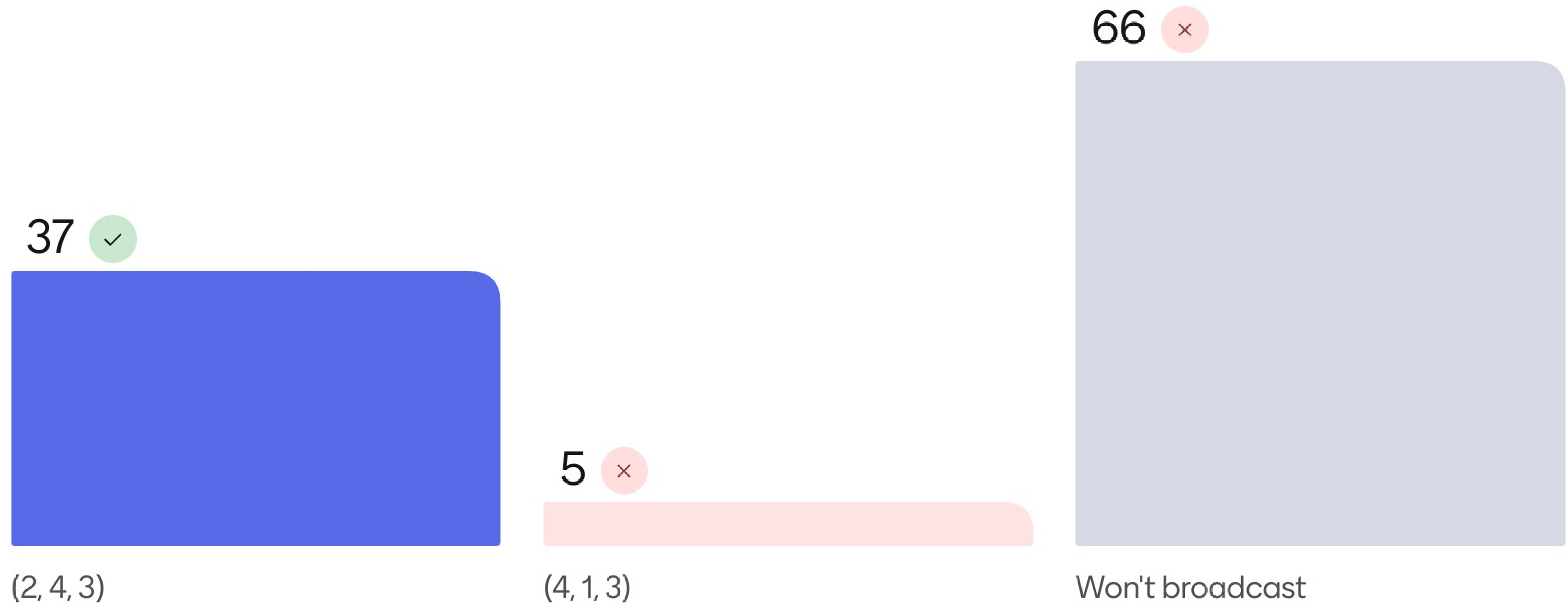
4	1
5	2
6	3

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Let $a.shape = (2, 3)$ and $b.shape = (3,)$. What is the broadcasted shape of $a + b$?



Let $a.shape = (2, 1, 3)$ and $b.shape = (4, 1)$. What is the broadcasted shape of $a + b$?



Let $a.shape = (2, 1, 3)$ and $b.shape = (8, 4, 1)$. What is the broadcasted shape of $a + b$?

25 ✖



(8, 4, 3)

0 ✖



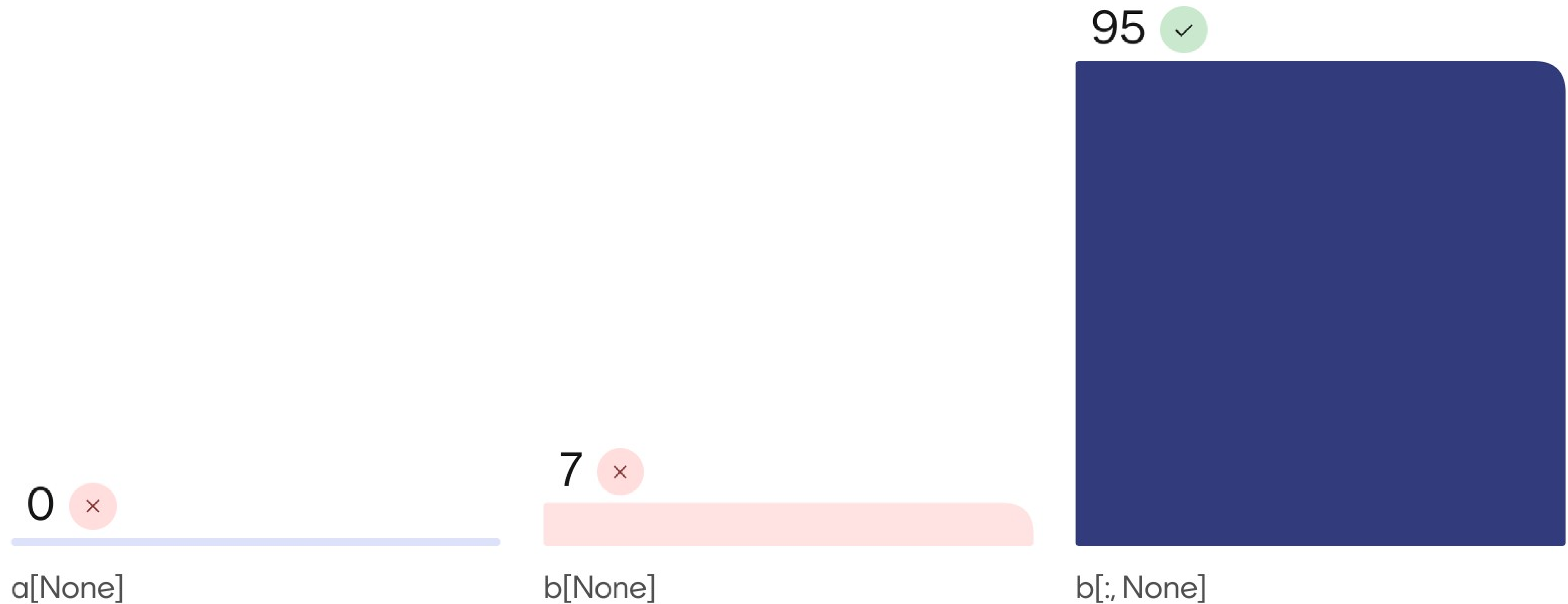
(2, 8, 4, 3)

82 ✔

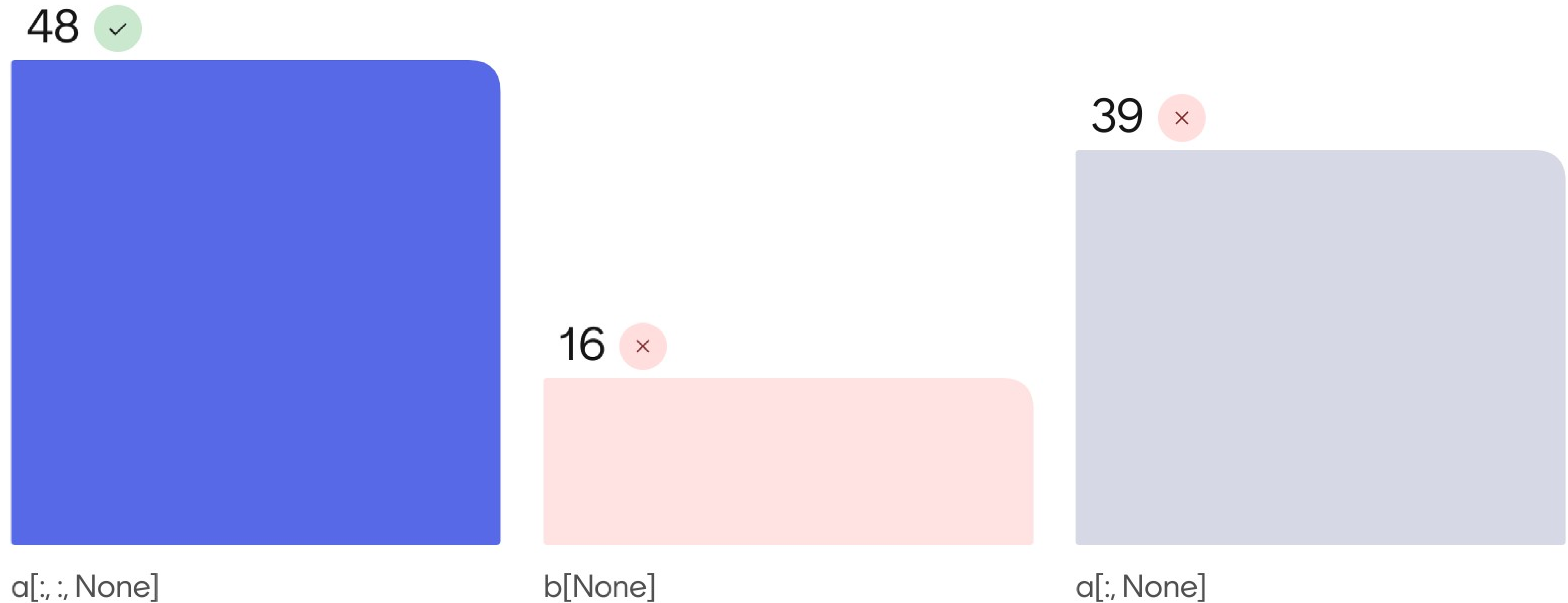


Won't broadcast

Let $a.shape = (2, 3)$ and $b.shape = (2,)$. What will make $a + b$ broadcastable?



Let $a.shape = (2, 3)$ and $b.shape = (2,)$. What will $a + b$ broadcastable?



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